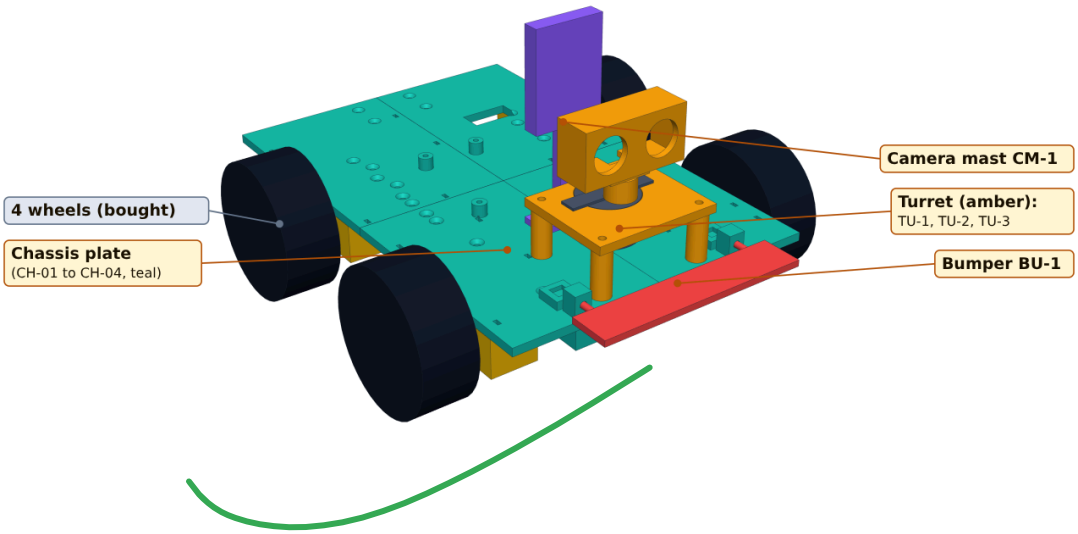
T-Minus_6 · Printed Parts Assembly Guide

How to put the 3D-printed parts together, step by step, with a labelled picture for every step. Team D NOVAS · NASA Space Apps 2026 Toronto.

The finished robot: every printed part in place



Read this first. The parts were designed for 2 wheels and a ball caster. You now have **4 wheels**. Four things change: (1) print 2 more saddles, (2) drill 4 holes for the front saddles (template on page 8), (3) use CH-10 only as a bridge with no marble, (4) CH-11 is not used. Everything else is exactly as designed.

Parts are shown from the STL files. Grey parts are already built in the earlier steps. Slate-grey labels mark parts you bought.

How to use this guide

- Each step has a picture, a list of what you need, what to do, and a check before the next step.
- Pictures marked *“view from underneath”* show the plate flipped over, so you can see parts that are under it. On your table the plate lies **top face up**. Left and right swap in those pictures.
- Part codes are printed in the file names: CH = chassis, TU = turret, CM = camera, BU = bumper, LA = launcher (asteroid tower), MC = controller box, ES = Earth stand.
- Printed parts are labelled in **yellow boxes**. Bought parts are labelled in **grey boxes**.

COLOUR CODE

Chassis (CH)	Turret (TU)	Camera (CM)	Bumper (BU)
Tower (LA)	Weather box (MC)	Earth stand (ES)	Bought part

BUILD ORDER AT A GLANCE

Step	What	Parts
1	Join tiles with straps	CH-01 to CH-05
2	Power bank cradle, join the two halves	CH-06, CH-07
3	Rear bridge (no caster ball)	CH-10
4	Saddles, motors, wheels	CH-08 ×2, CH-09 ×2
5	Turret	TU-1, TU-2, TU-3
6	Camera mast and bumper	CM-1, BU-1
7	Electronics on the plate	(bought parts)
8	Asteroid tower	LA-1, LA-2, LA-3
9	Weather-station box	MC-1, MC-2
10	Earth stand and obstacles	ES-1, ES-2

TOOLS

- Phillips screwdriver (small), pliers, scissors for zip ties
- Hand drill with a 3.4 mm bit (4 new holes)
- Calipers or a ruler, pencil, double-sided tape, CA (super) glue, a few books to prop up the plate

Screws and small parts

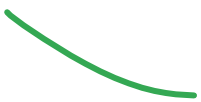
Part	How many	Used for
M3 × 10 flat-head (countersunk)	**28** (24 in the old list, +4 for the extra saddles)	Straps 8, cradle ears 8, saddles 8, CH-10 flange 4
M3 × 30 pan-head	4	Stepper cradle legs (TU-1)
M3 × 8 pan-head + washer	2	28BYJ-48 ears
M3 × 10 pan-head	10	Camera mast 2, tower base 4, controller lid 4
M3 × 6 pan-head	8	UNO Q 4, UNO R3 4
M2.5 × 8	4	LCD onto the lid
Zip ties 2.5 mm	about 24 + 4	Modules and motors
CA glue	1 tube	Hub, sensor holder, servo horns

The old list had exactly 4 spare M3×10 flat-head screws. The 4 extra saddle screws use them all up. Buy a few more.

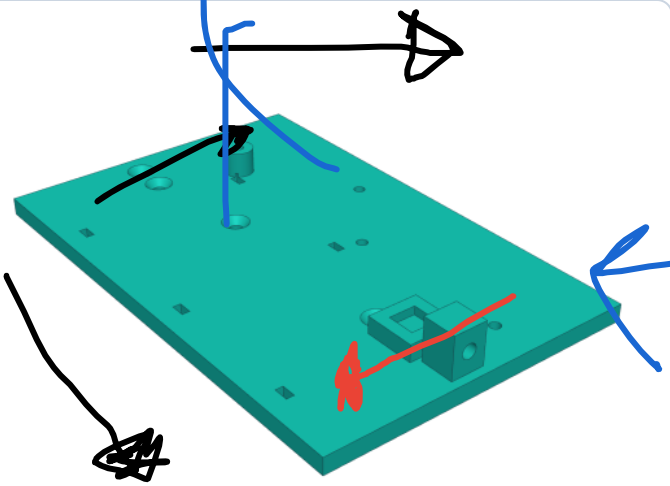
PRINT LIST (WHAT TO PRINT NOW)

Part	Print
All 23 files you already have	1 each, except CH-05 ×2 and ES-2 ×2
CH-08 Motor saddle LEFT	**1 more** (so 2 in total)
CH-09 Motor saddle RIGHT	**1 more** (so 2 in total)
CH-11 Caster ring	Not needed (spare)

Saddle print orientation: flip it upside down so the ears lie on the bed and the cavity opens downward (no supports).



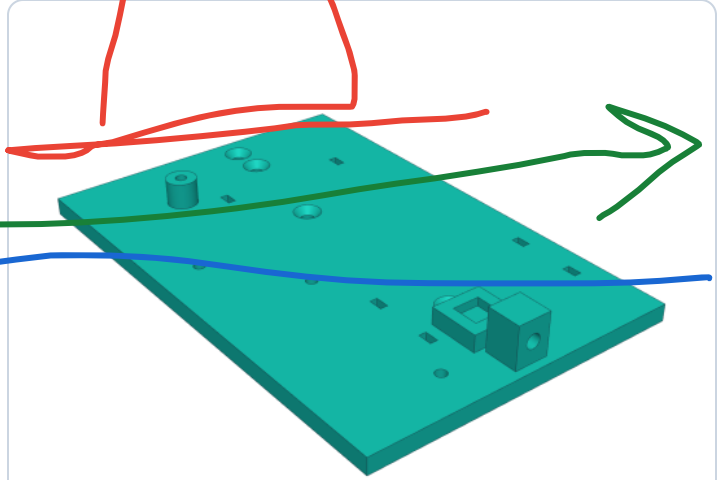
The 23 parts (25 pieces) and what each one is



CH-01 **Plate tile front-left**

Qty: **1** Size mm: 75 × 114 × 14 Step: **1**

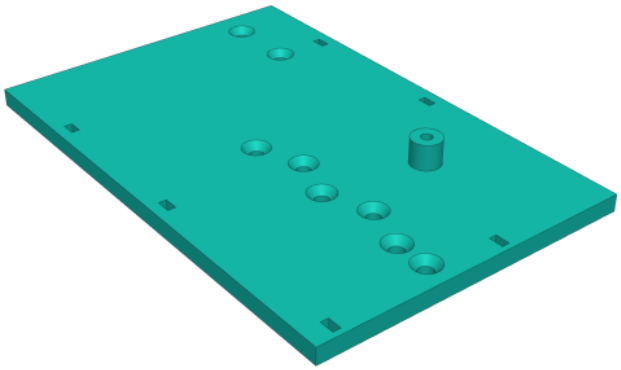
Front-left quarter of the base plate. Has the left bumper guide block and switch seat.



CH-02 **Plate tile front-right**

Qty: **1** Size mm: 75 × 114 × 14 Step: **1**

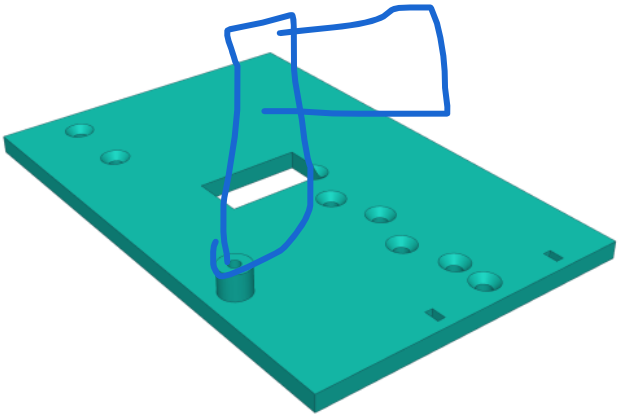
Front-right quarter of the plate. Right bumper guide block and switch seat.



CH-03 **Plate tile rear-left**

Qty: **1** Size mm: 75 × 114 × 10 Step: **1**

Rear-left quarter. Has two UNO Q standoff posts.

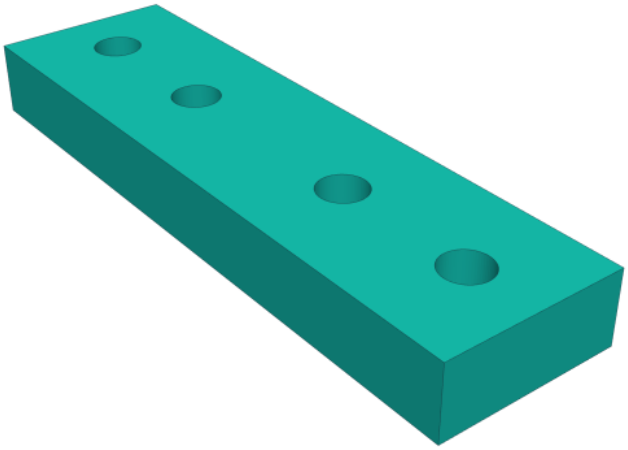


CH-04 **Plate tile rear-right**

Qty: **1** Size mm: 75 × 114 × 10 Step: **1**

Rear-right quarter. Has two UNO Q standoff posts and the cable slot.

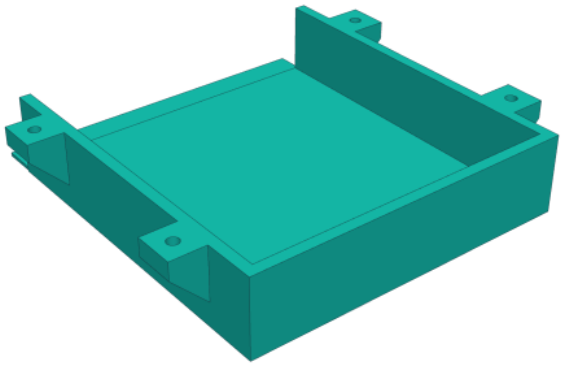
The 23 parts (25 pieces) and what each one is



CH-05 **Y-seam strap**

Qty: **2** Size mm: 10 × 36 × 4 Step: **1**

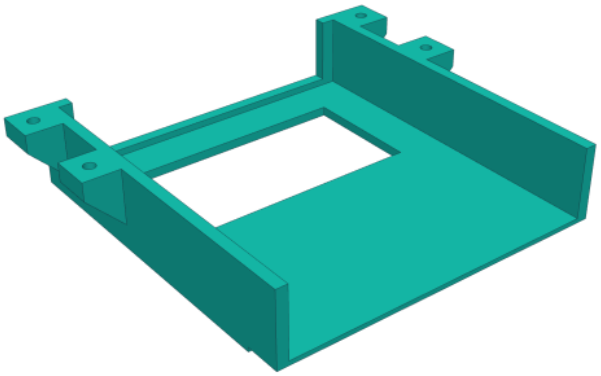
Flat strip that joins a front tile to a rear tile. One on the left seam, one on the right.



CH-06 **Bank cradle, front half**

Qty: **1** Size mm: 90.5 × 85.6 × 17.5 Step: **2**

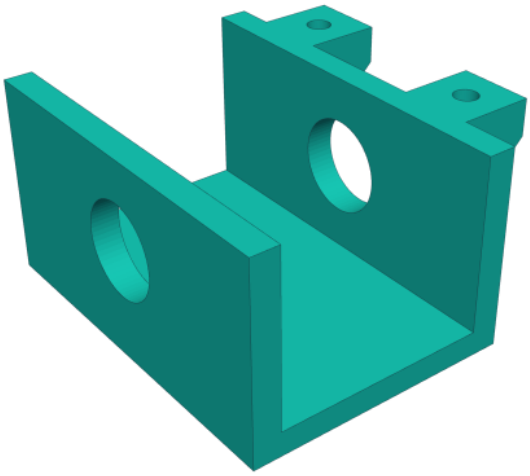
Front half of the tray under the plate that holds the power bank. Has a tongue that goes under the rear half.



CH-07 **Bank cradle, rear half**

Qty: **1** Size mm: 90.5 × 77.8 × 17.5 Step: **2**

Rear half of the tray. Has the port window (rear wall) and the button slot in the floor.



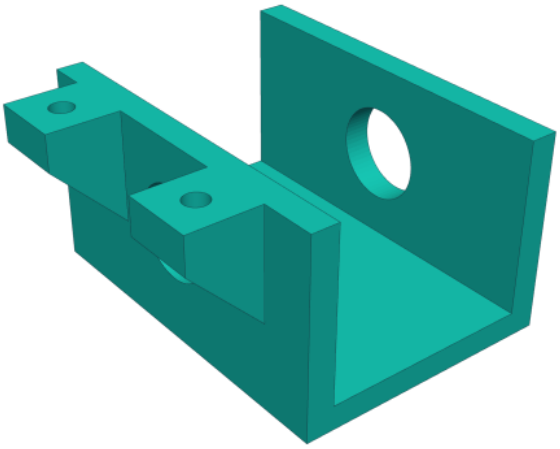
CH-08 **Motor saddle LEFT**

Qty: **2 (print 1 more)** Size mm: 36.8 × 40 × 21.5

Step: **4**

Clamp that holds one TT motor under the left side of the plate. You need two: front-left and rear-left.

The 23 parts (25 pieces) and what each one is

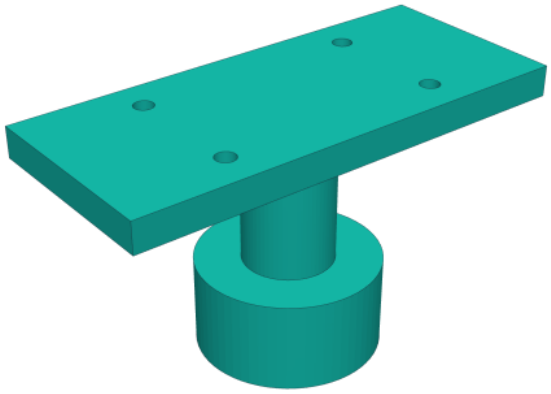


CH-09 **Motor saddle RIGHT**

Qty: **2 (print 1 more)** Size mm: 36.8 × 40 × 21.5

Step: **4**

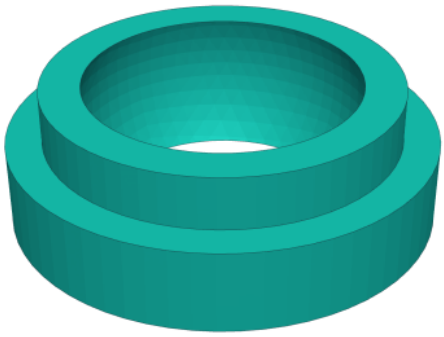
Same, for the right side: front-right and rear-right.



CH-10 **Caster body (upper)**

Qty: **1** Size mm: 60 × 26 × 34 Step: **3**

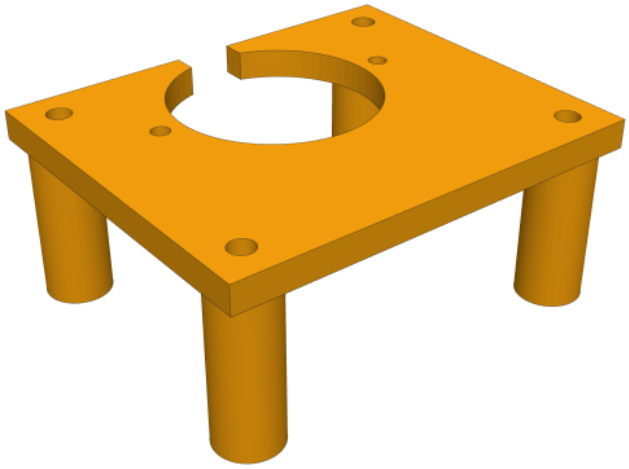
Was the ball-caster. With 4 wheels it is only used as the bridge that joins the rear tiles.



CH-11 **Caster ring (lower)**

Qty: **1 (spare)** Size mm: 24 × 24 × 7.7 Step: **not used**

Closes the caster cup under the marble. NOT used with 4 wheels. Keep it as a spare.



TU-1 **Stepper cradle**

Qty: **1** Size mm: 60 × 50 × 26 Step: **5**

Four-legged table that holds the 28BYJ-48 stepper motor above the plate.

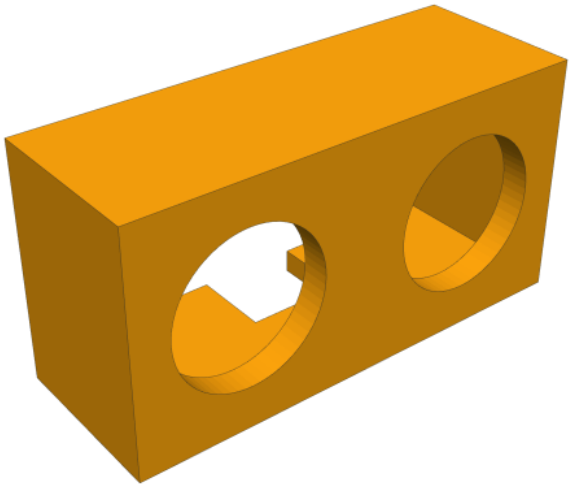
The 23 parts (25 pieces) and what each one is



TU-2 **Sensor hub**

Qty: **1** Size mm: 20 × 14 × 13 Step: **5**

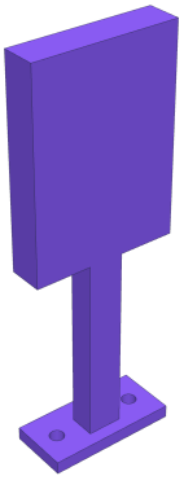
Fits on the stepper shaft and carries the sensor holder.



TU-3 **Sensor holder (HC-SR04)**

Qty: **1** Size mm: 50.6 × 18.5 × 25.8 Step: **5**

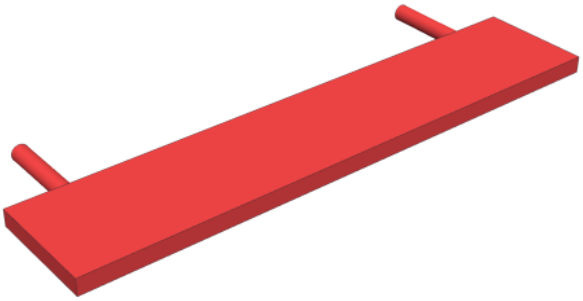
Small box that holds the HC-SR04 distance sensor. Two round holes for its eyes.



CM-1 **Camera mast**

Qty: **1** Size mm: 33.6 × 14 × 93 Step: **6**

Tall post with a tray that holds the ESP32-CAM, camera looking forward.

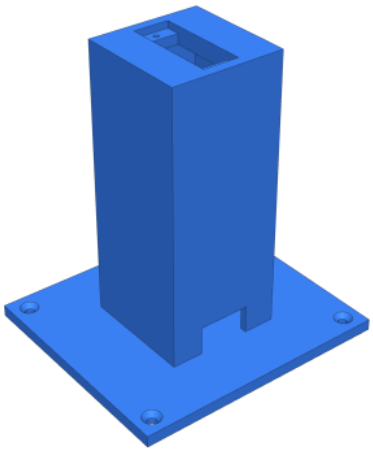


BU-1 **Front bumper**

Qty: **1** Size mm: 100 × 34 × 3 Step: **6**

Floating bar with two pins. When it bumps, a pin presses a switch.

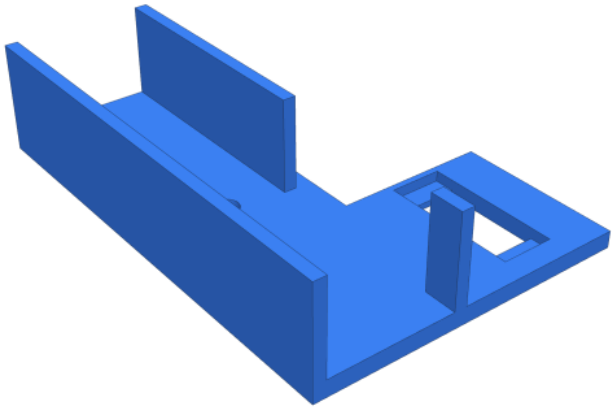
The 23 parts (25 pieces) and what each one is



LA-1 **Launcher base + tower**

Qty: **1** Size mm: 90 × 80 × 100 Step: **8**

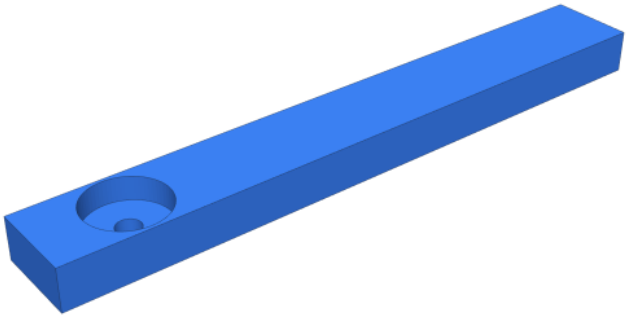
Base with a hollow pillar. The pan servo sits in the top.



LA-2 **Launcher head (hopper)**

Qty: **1** Size mm: 76 × 110 × 26 Step: **8**

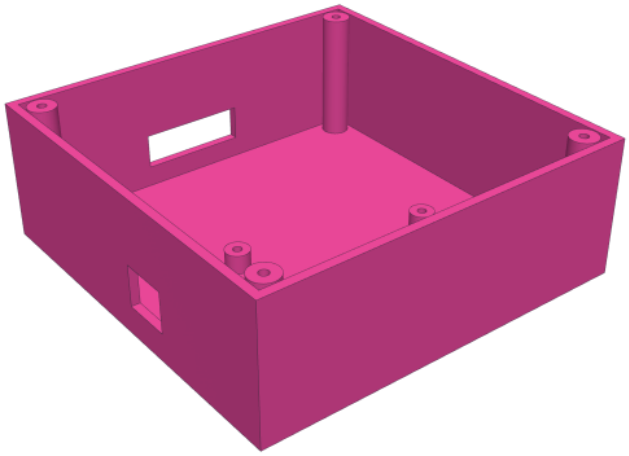
Channel for the paper ball plus a deck for the push servo. Turns on the pan servo.



LA-3 **Pusher arm**

Qty: **1** Size mm: 60 × 8 × 4 Step: **8**

Arm on the push servo horn. Swings and pushes the ball out.

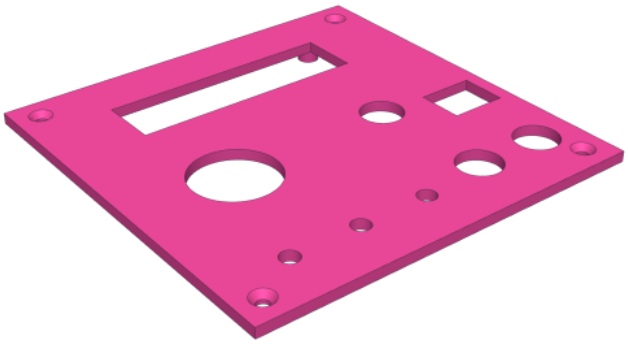


MC-1 **Controller base (weather box)**

Qty: **1** Size mm: 112 × 112 × 40 Step: **9**

Open box for the Arduino UNO R3 of the weather station.

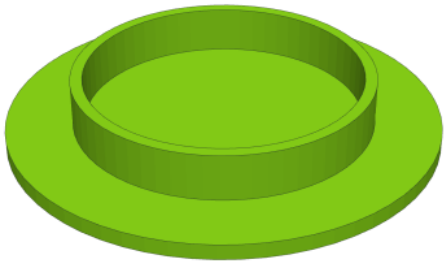
The 23 parts (25 pieces) and what each one is



MC-2 **Controller lid**

Qty: **1** Size mm: 112 × 112 × 9 Step: **9**

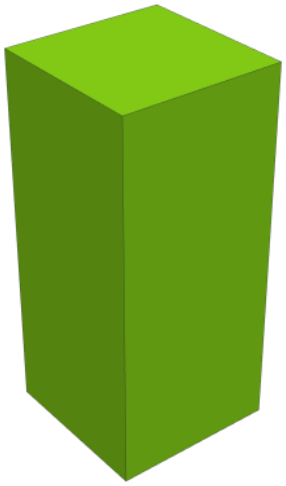
Lid with the LCD window, LED holes, buzzer holes and DHT11 window.



ES-1 **Earth stand**

Qty: **1** Size mm: 110 × 110 × 16 Step: **10**

Round stand with a ring that holds the 70 mm foam Earth ball.



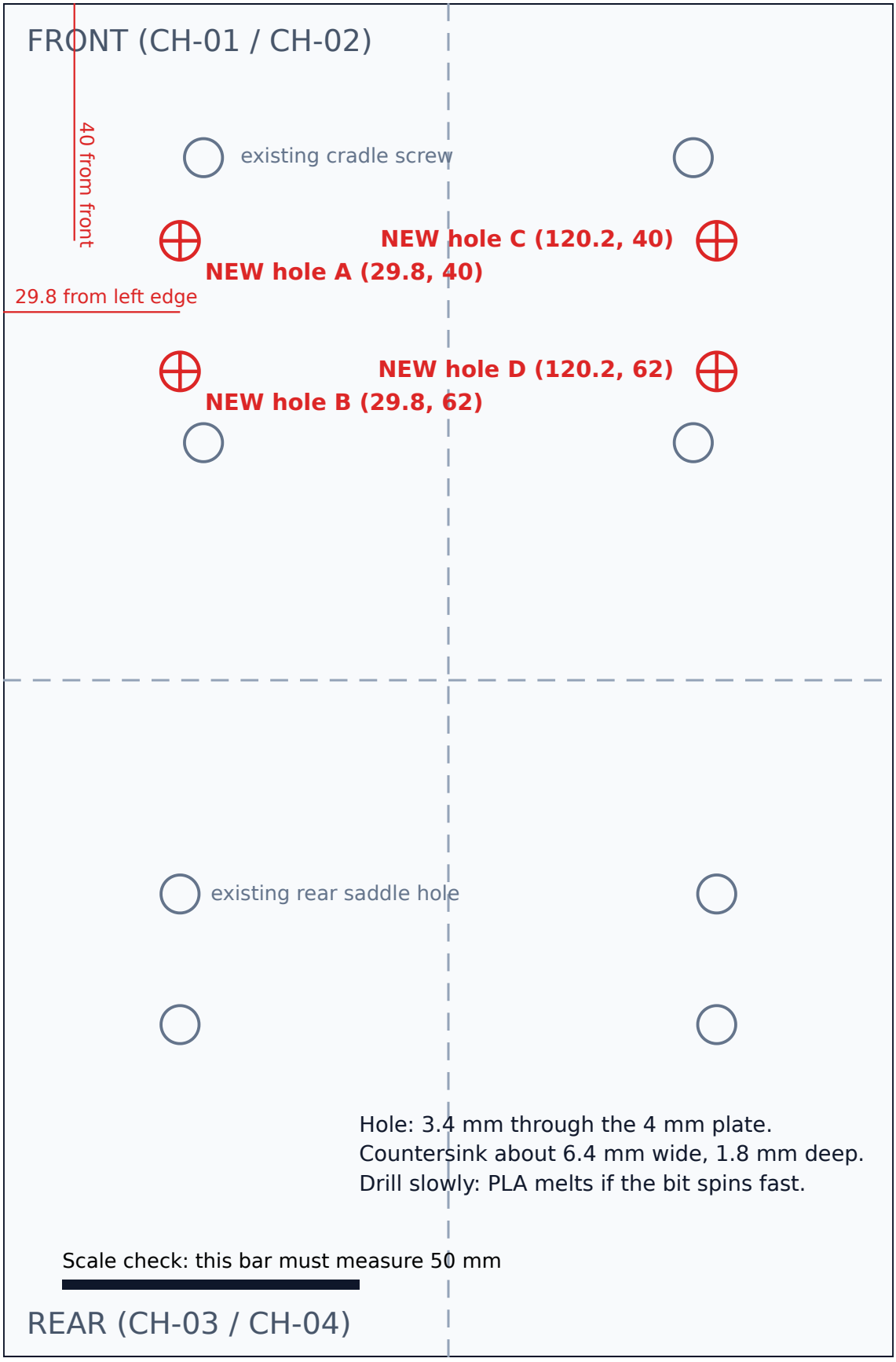
ES-2 **Debris obstacle**

Qty: **2** Size mm: 40 × 40 × 90 Step: **10**

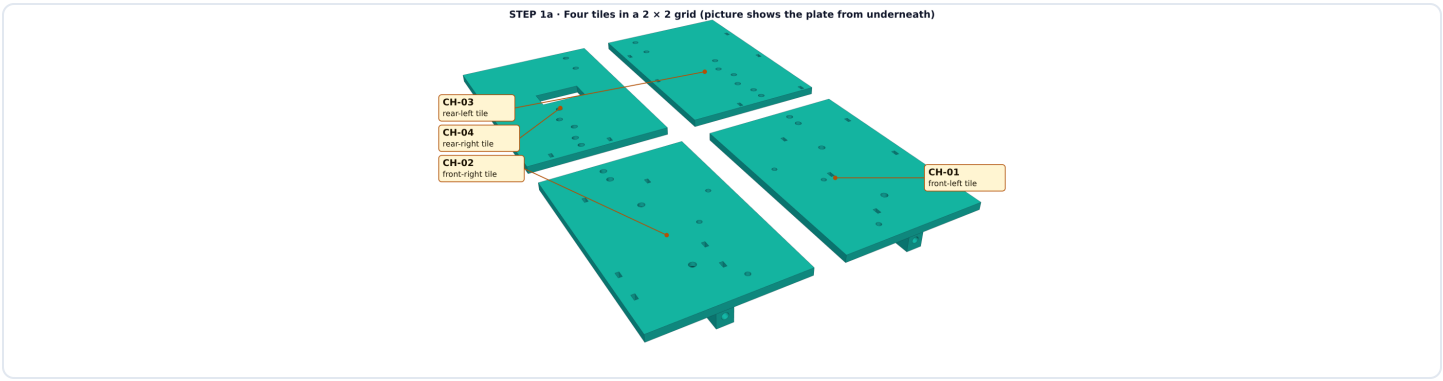
Hollow post used as a fixed obstacle on the board.

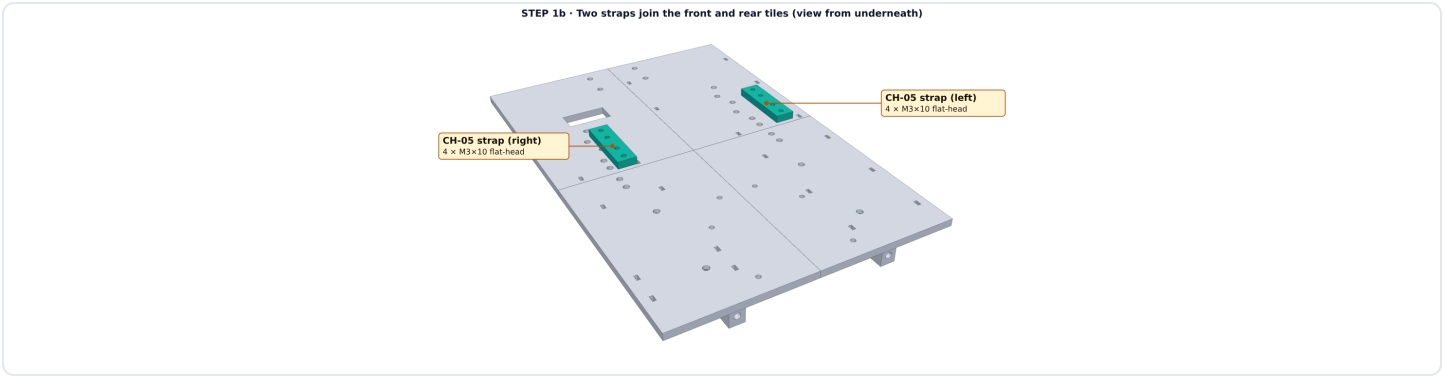
Drill template: 4 new holes for the front saddles

Print at 100% (Actual size) and check the 50 mm bar. Top face of the plate, front edge at the top. Best method: hold a saddle under the plate and mark through its ear holes.



1 Join the four tiles into two halves (CH-01 to CH-05)





YOU NEED

- CH-01, CH-02, CH-03, CH-04
- 2 × CH-05 straps
- 8 × M3×10 flat-head (countersunk) screws
- Phillips screwdriver

CHECK BEFORE MOVING ON

- Every screw head is flush with the top face (no head sticking up).
- The seam between front and rear tile is closed with no gap.
- Left half and right half are each 75 mm wide and 228 mm long.

DO THIS

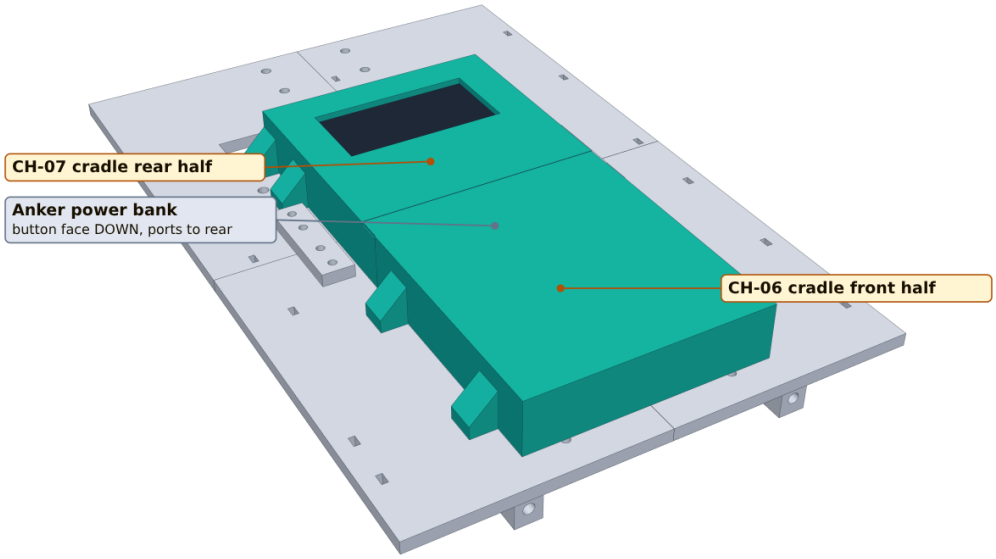
1. Put the two straps (CH-05) flat on the table. Space them like the picture: one near the left edge, one near the right edge (about 90 mm apart).
2. Lay **CH-01** (front-left) and **CH-03** (rear-left) on top of the left strap, **top face up**, seam to seam. The strap is hidden under the seam.
3. Look straight down through the four countersunk holes at the seam: you see the holes of the strap underneath. Move the strap until they line up.
4. Screw 4 × M3×10 flat-head screws down through the plate into the strap. The strap threads are self-cutting, so push firmly and turn slowly. Stop when the heads are flush.
5. Do the same on the right with **CH-02** (front-right), **CH-04** (rear-right) and the second strap.
6. You now have two long halves (left half and right half). Do not join them yet.

Tip: The pictures show the plate seen from underneath so you can see what goes where. On your table the plate lies top face up. Left and right swap in the picture.

Careful: Do not over-tighten. PLA can crack. Stop when the head touches the plate.

2 Power bank in the cradle, then join the halves (CH-06, CH-07)

STEP 2 · Power bank in the cradle, plate screwed on top (view from underneath)



YOU NEED

- CH-06 and CH-07 (cradle halves)
- Anker power bank
- 8 × M3×10 flat-head screws
- Two stacks of books (about 4 cm tall)

CHECK BEFORE MOVING ON

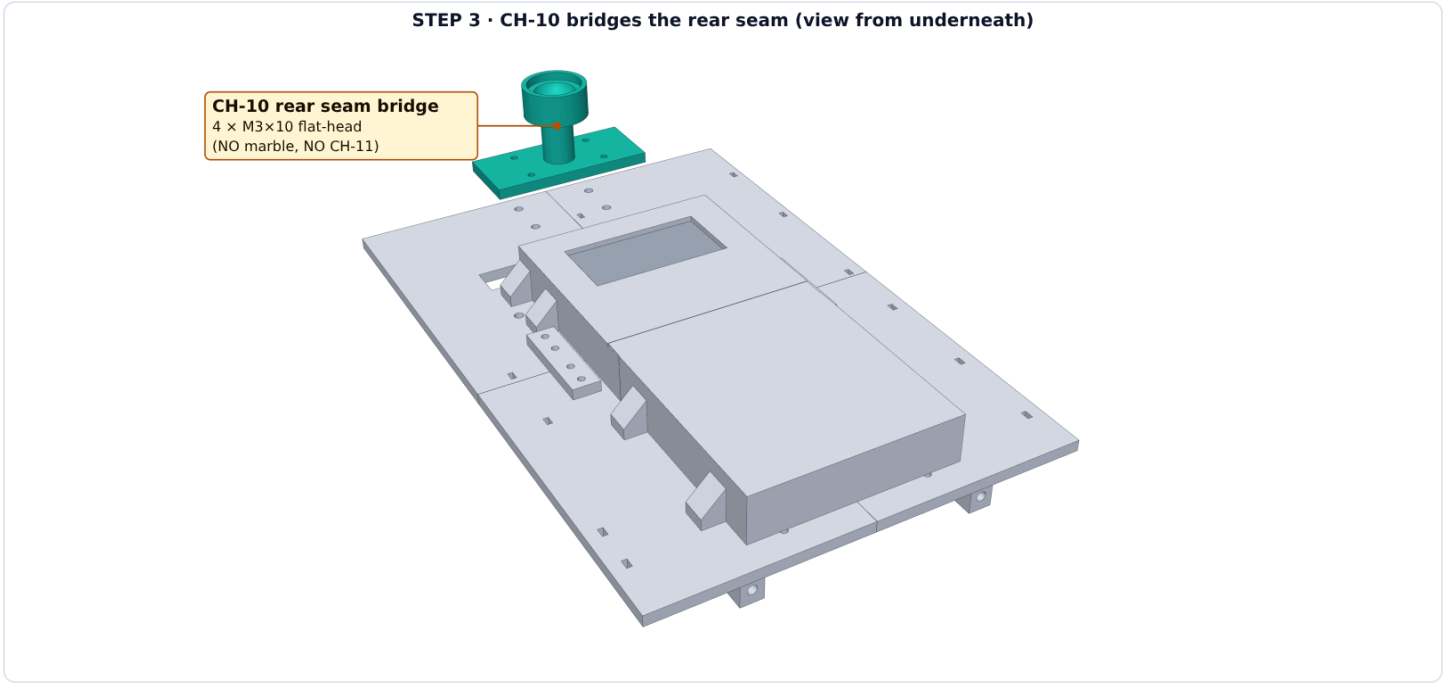
- The plate is now one stiff piece.
- Both USB ports of the power bank line up with the rear window.
- The power button is under the slot (you can press it from below).

DO THIS

1. Put the front cradle half (**CH-06**) on the table with its open side up. The small tongue sticks out at its rear end.
2. Slide the rear half (**CH-07**) against it so the front edge of CH-07 overlaps the tongue of CH-06. The two halves now form one long tray.
3. Lay the power bank in the tray with its **button facing DOWN** (the button goes into the long slot in the floor of CH-07) and the **ports toward the rear** (they face the port window).
4. Lower the left half-plate and the right half-plate onto the cradle, **top face up**, with the middle seam (x = 75 mm) over the middle of the tray. Rest the plate edges on the book stacks if it wobbles.
5. Each cradle half has 4 ears (small flat tabs). Look through the countersunk holes at x ≈ 34 and x ≈ 116: you see the ear holes. Line them up.
6. Screw the 8 ear screws (M3×10 flat-head) down through the plate into the ears.

Tip: If the plate rocks, put the books under the rear edge. Do not press the plate onto the bank: the cradle was sized so the plate just touches the top.

3 Rear bridge CH-10 (no caster ball)



YOU NEED

- CH-10
- 4 × M3×10 flat-head screws

CHECK BEFORE MOVING ON

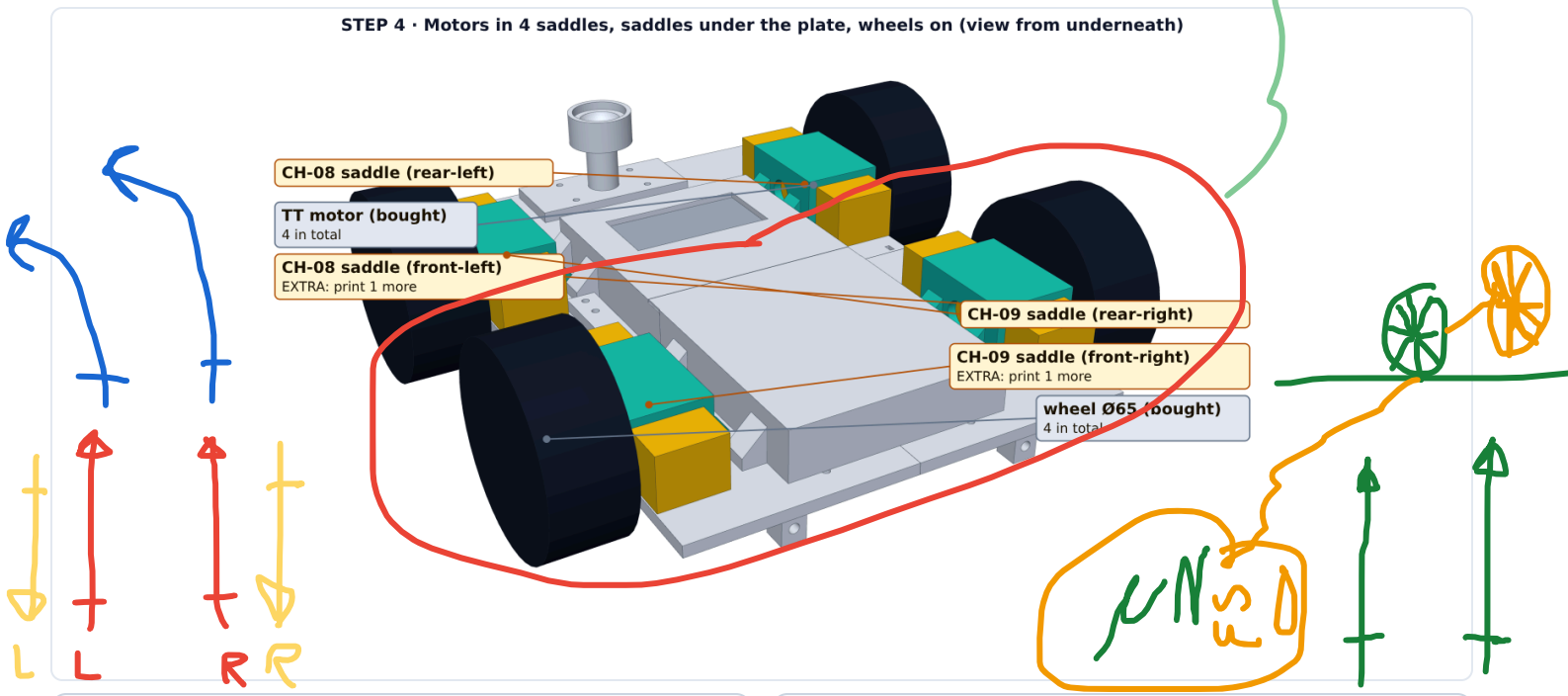
- The two rear tiles cannot flex at the seam any more.
- The cup hangs straight down, in the middle of the rear edge.

DO THIS

1. Slide the whole plate toward the table edge so the **rear end hangs over the edge** (the cup of CH-10 is 34 mm tall and needs space below).
2. From below, hold CH-10 against the underside of the rear tiles. Its flat flange (60 × 26) sits across the seam at x = 75 mm, y = 199 to 225 mm.
3. Look through the 4 countersunk holes near the rear edge: line up the flange holes. Screw 4 × M3×10 down through the plate into the flange.
4. **Do not** fit CH-11 or a marble. With 4 wheels the robot rolls on the wheels; the cup just hangs about 8 mm above the floor and acts as a skid.

Careful: If the marble touches the floor, the robot will rock on 5 points. Leave it out.

4 Saddles, motors and wheels (CH-08, CH-09)



YOU NEED

- 4 saddles: 2 × CH-08 (left) and 2 × CH-09 (right) **(you must print 1 more of each)**
- 4 TT gear motors, 4 wheels
- 8 × M3×10 flat-head screws
- Drill with a 3.4 mm bit (for the 4 new front holes, see the next page)

CHECK BEFORE MOVING ON

- All four wheels touch the table at the same time.
- Wheel axle height is 32.5 mm, so the plate bottom floats 10 mm above wheel centre line: the robot should sit level.
- Rear wheel centre is 110 mm behind the front wheel centre.

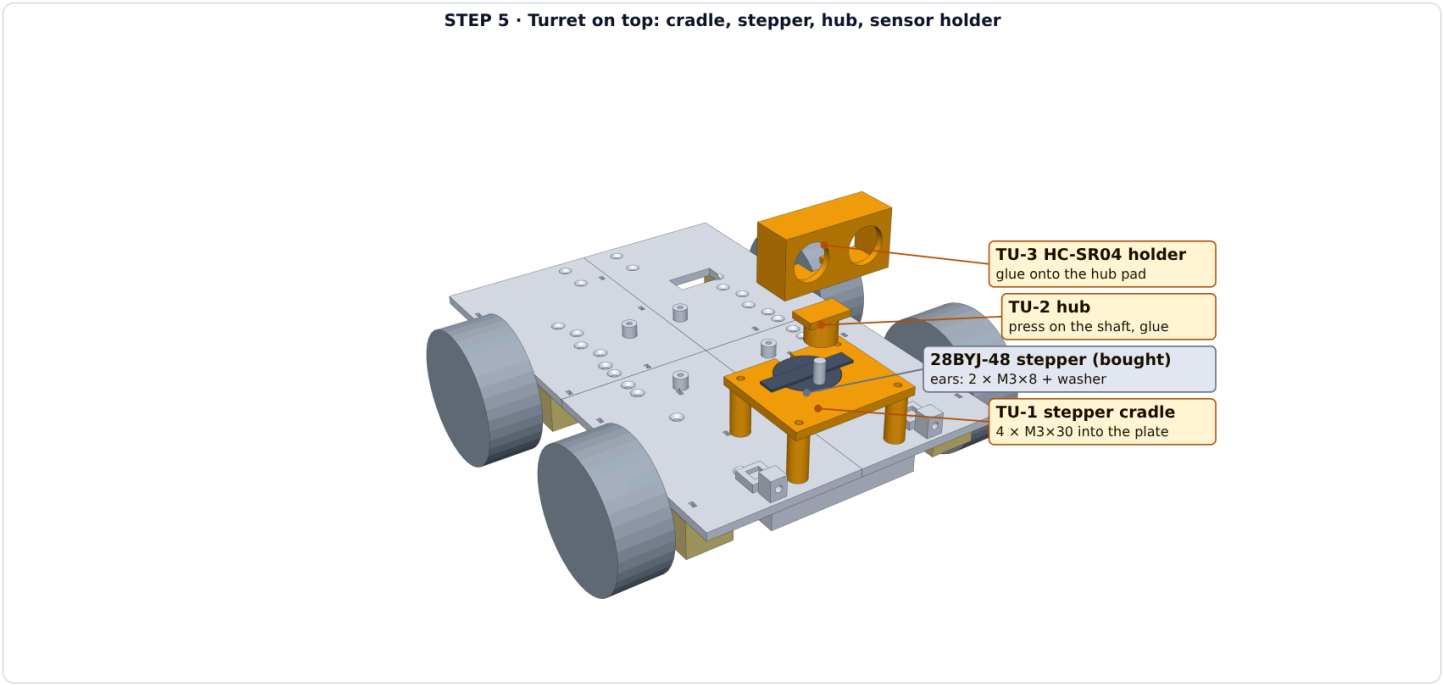
DO THIS

1. **Rear saddles** use the holes that are already in CH-03 and CH-04 (x = 29.8 and 120.2, y = 150 and 172).
2. **Front saddles**: first make the 4 new holes in CH-01 and CH-02 with the drill template on the next page.
3. Slide a TT motor into a saddle from the front or back. The motor lies lengthways (70 mm long) with its axle going through the round hole in both side walls. The axle sticks out on the outer side.
4. Hold the saddle under the plate edge with the ears flat against the underside. Line the ear holes with the countersunk holes. Screw 2 × M3×10 per saddle.
5. Add a zip tie around each motor and saddle for extra grip.
6. Push a wheel onto each axle. Check the left motors turn the same way as each other (same direction of wire).

Tip: Wire the two motors on the same side in parallel: they always turn the same way. The robot turns like a tank.

Careful: Left saddle and right saddle are different shapes (mirror images). Check the code on the part (CH-08 left, CH-09 right).

5 Turret: stepper cradle, hub and sensor holder (TU-1, TU-2, TU-3)



YOU NEED

- TU-1, TU-2, TU-3
- 28BYJ-48 stepper motor
- 4 × M3×30 pan-head screws, 2 × M3×8 + washers
- HC-SR04 sensor, CA (super) glue

CHECK BEFORE MOVING ON

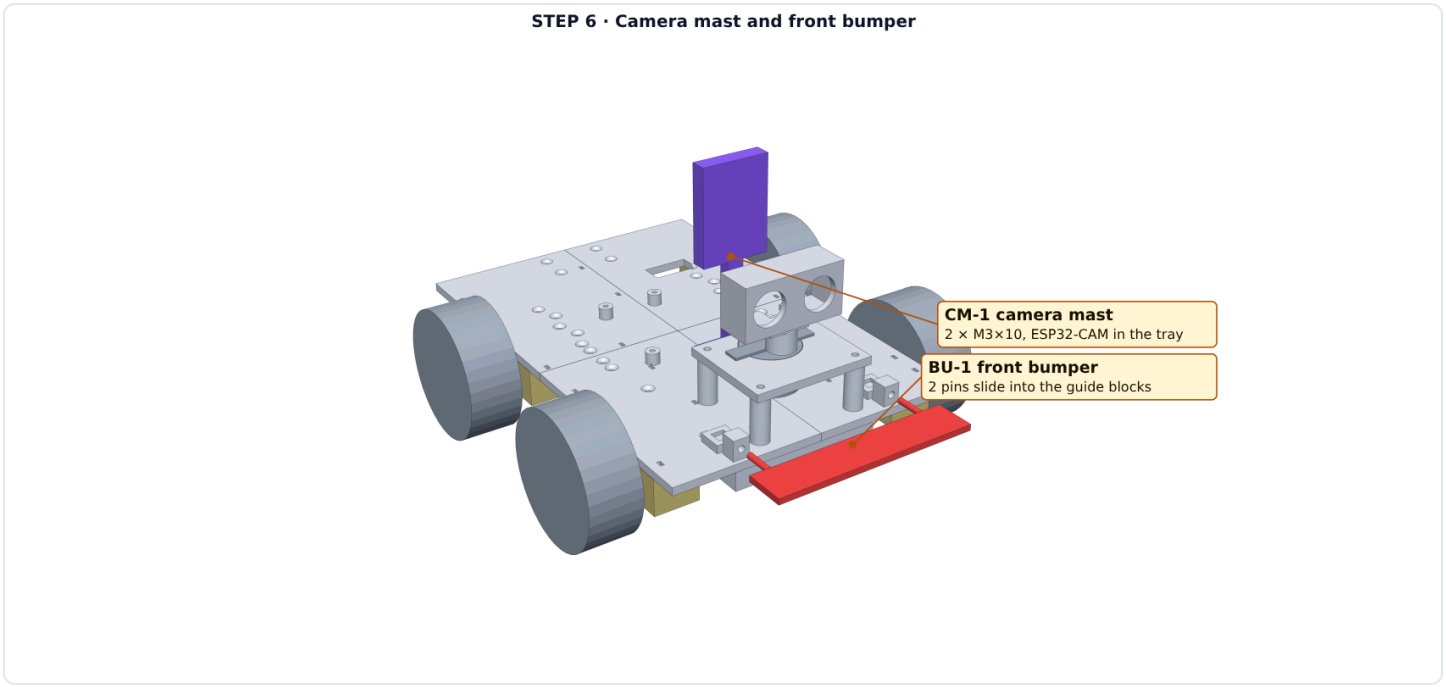
- The hub turns freely with the stepper.
- The sensor looks straight forward when the stepper is at its centre position.

DO THIS

1. Flip the robot top face up. Put **TU-1** on the plate at the front (x 45 to 105, y 5 to 55). The four legs sit over the four pilot holes.
2. Screw 4 × M3×30 down through the corners of TU-1, through the legs, into the plate.
3. Drop the stepper into the big round hole of TU-1: the round can goes down through the hole and the two ears rest on the top. The shaft points up.
4. Fix the ears with 2 × M3×8 screws and washers (holes are 35 mm apart). Put the cable through the little notch at the rear of the hole.
5. Press **TU-2** (hub) onto the shaft. The hole is D-shaped: turn it until the flats match. Add a tiny drop of CA glue.
6. Glue **TU-3** (sensor holder) onto the flat pad on top of the hub, round holes facing forward.
7. Slide the HC-SR04 into TU-3 from the back so its two silver eyes poke out through the round holes.

Tip: Put the stepper cable in the notch first, or the hub will pinch it.

6 Camera mast and front bumper (CM-1, BU-1)



YOU NEED

- CM-1, BU-1
- 2 × M3×10 pan-head screws
- ESP32-CAM board (or a phone)
- 2 small switches (tact or micro, about 6 mm)

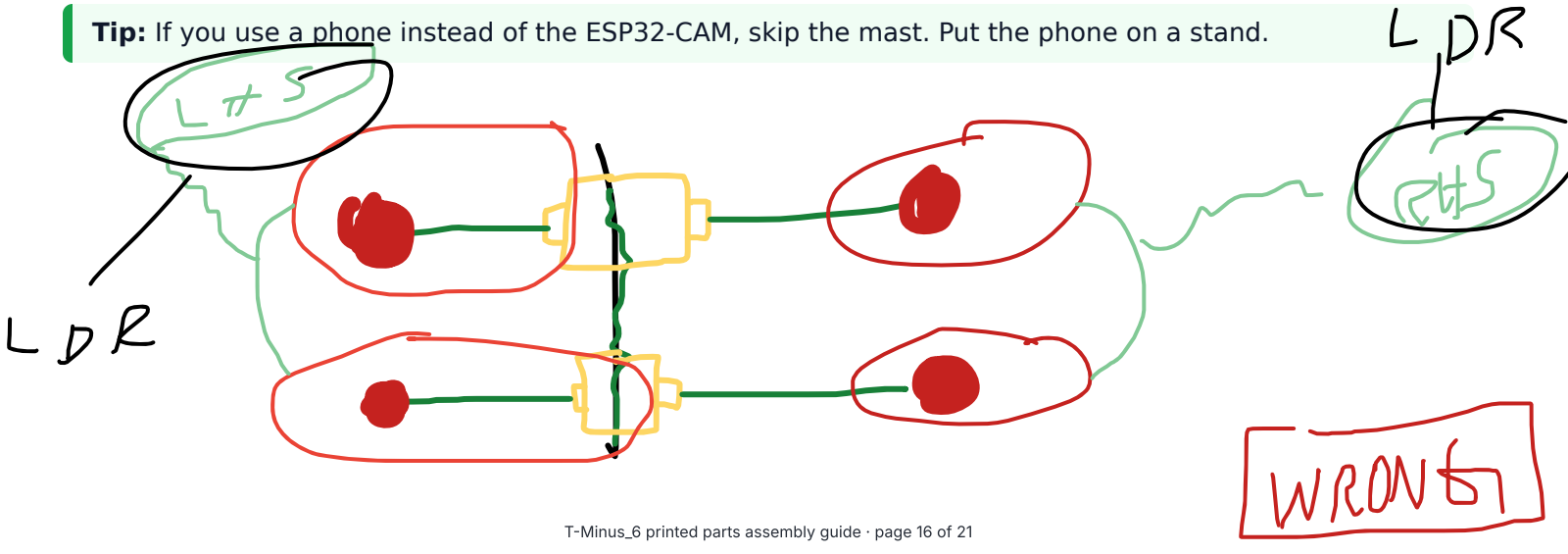
CHECK BEFORE MOVING ON

- Bumper slides freely, no rubbing.
- The camera sees forward over the sensor.

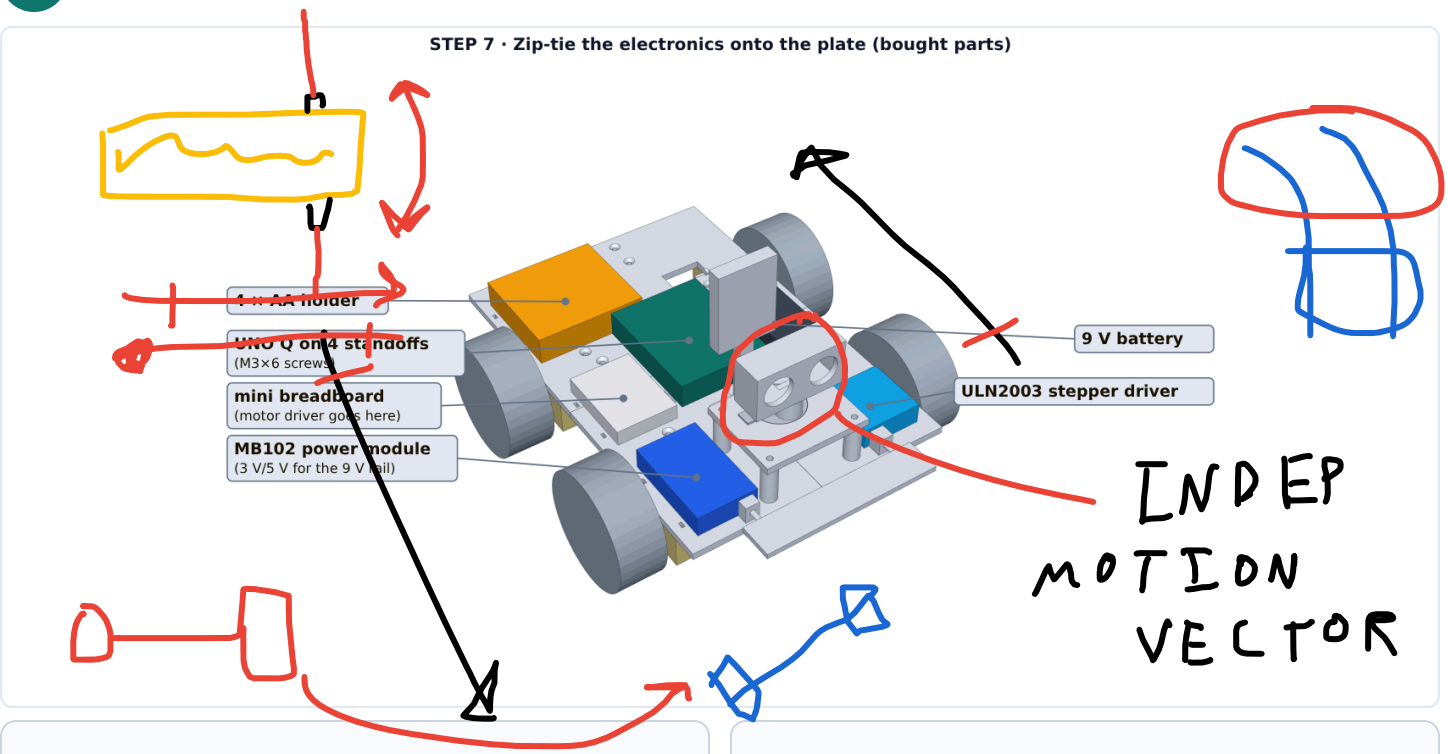
DO THIS

1. Put **CM-1** on the rear of the plate (x 58 to 92, y 62 to 76). Screw 2 × M3×10 at the two base holes.
2. Lay the ESP32-CAM in the tray: lens in the 14 × 14 mm window, USB side down. A rubber band or tape holds it.
3. Put a small switch in each of the two **switch seats** on the front of the plate (x = 29 and x = 109, y = 11).
4. Slide the two pins of **BU-1** into the two round holes of the guide blocks (at x = 35 and x = 115). The bar floats 14 mm in front of the plate.
5. Check: push the bumper straight back. It must press both switches and spring back.

Tip: If you use a phone instead of the ESP32-CAM, skip the mast. Put the phone on a stand.



7 Electronics onto the plate (bought parts, zip ties)



YOU NEED

- MB102 power module, ULN2003 driver, mini breadboard, 9 V battery, 4 × AA holder, UNO Q (×1, second one on the tower)
- About 24 zip ties (2.5 mm)
- 4 + 4 × M3×6 screws for the board standoffs

CHECK BEFORE MOVING ON

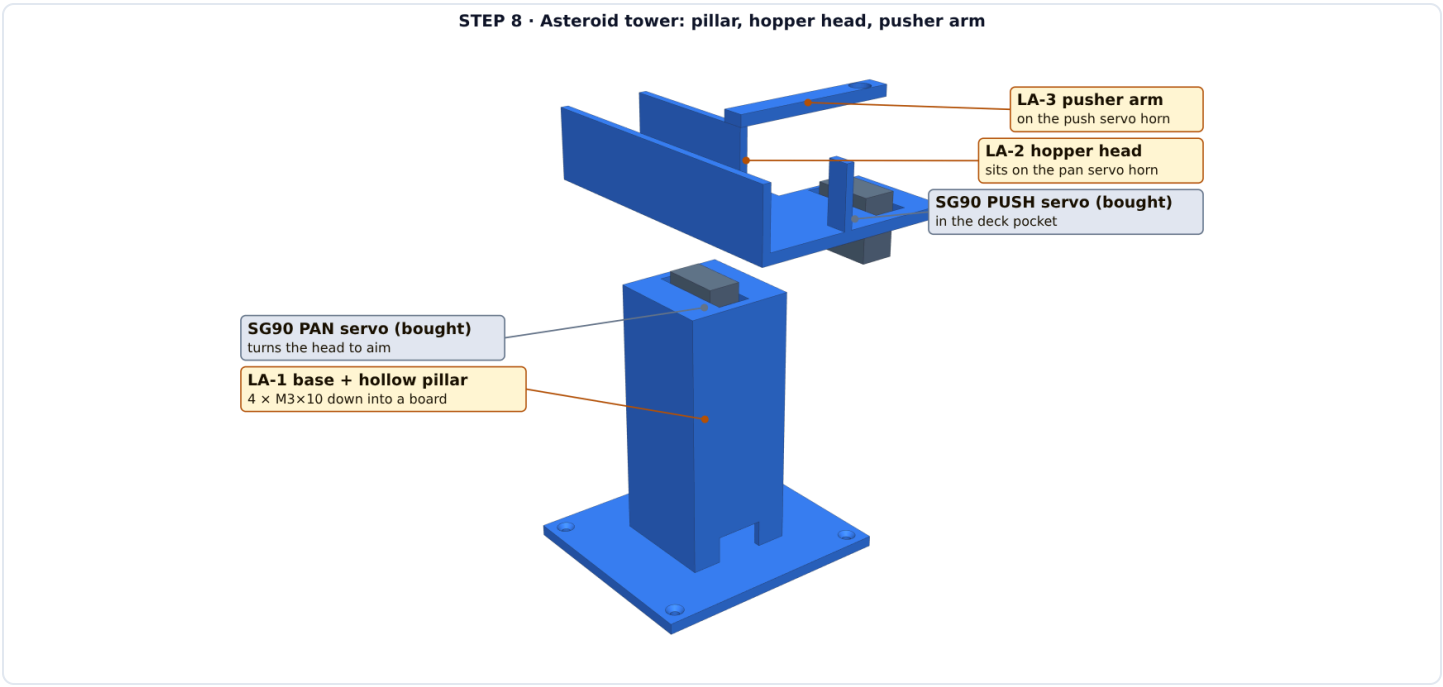
- Nothing touches the wheels or the turret.
- The UNO Q boards sit on the posts, not directly on the plate.

DO THIS

1. Zip-tie each module into its slots: the plate has pairs of small slots around every place. Pull the tie through the slots and over the module.
2. Put the UNO Q on its four round posts (CH-03, CH-04) with 4 × M3×6 screws.
3. Put the motor driver (DRV8833) on the mini breadboard; that breadboard is tied down at the front-left.
4. Route cables up through the long slot behind the UNO Q so the power bank cables reach the boards.

> **Tip:** The colours in the picture are only stand-ins for the real modules, so the exact shapes will differ.

8 Asteroid tower (LA-1, LA-2, LA-3)



YOU NEED

- LA-1, LA-2, LA-3
- 2 × SG90 servos with horns and screws
- 4 × M3×10 screws
- CA glue

CHECK BEFORE MOVING ON

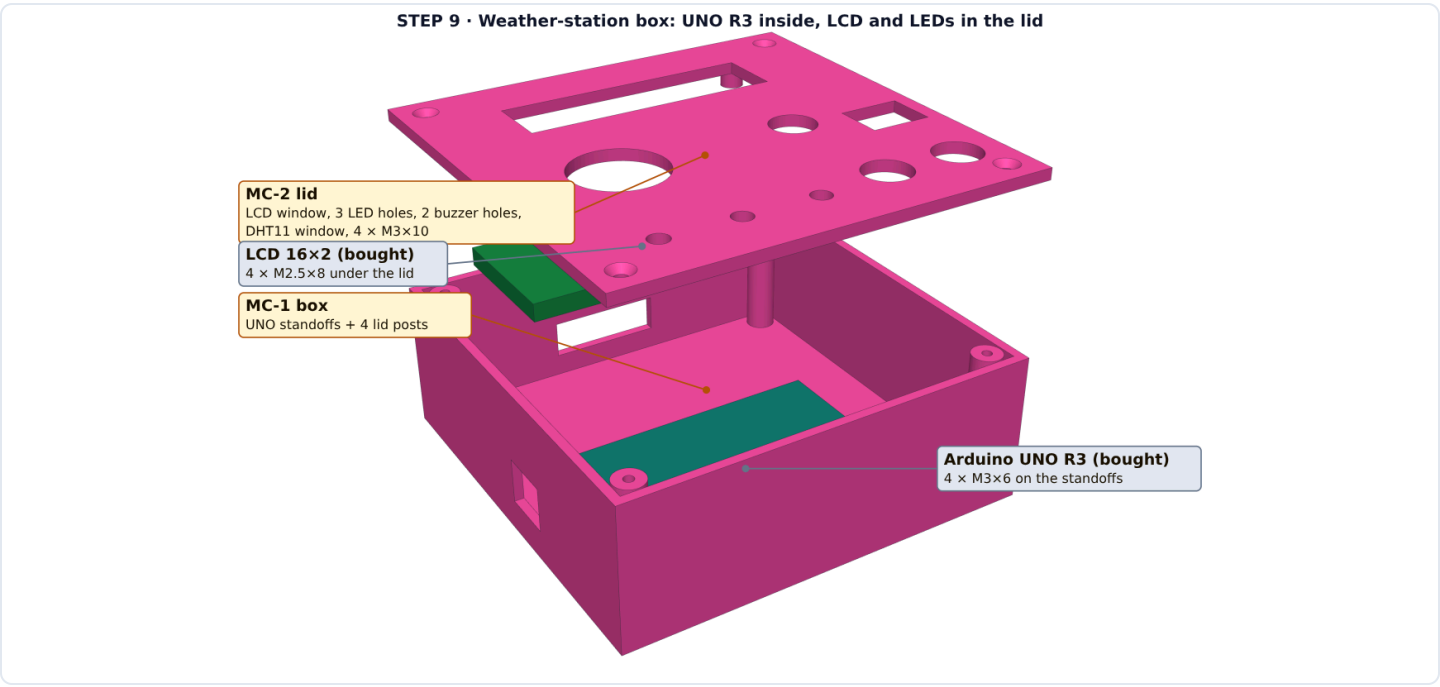
- Pan servo turns the head about 90° each side without hitting the pillar.
- Arm moves freely through the slot.

DO THIS

1. Screw **LA-1** to your mission board with 4 screws through the corner holes.
2. Drop the **pan servo** (SG90) into the rectangular pocket in the top of the pillar. Its ears sit in the recesses; fix with the two small ear screws.
3. Put the servo horn on the shaft. Glue or screw **LA-2** (hopper head) onto the horn: the round recess on its underside fits the horn hub.
4. Put the **push servo** in the pocket of the deck on LA-2 (the flat part beside the channel).
5. Fit **LA-3** (arm) on the push servo horn: the round recess on the underside of the arm fits the horn. One small screw holds it.
6. Check: the arm swings across the channel through the slot in the right wall and pushes a 25 mm paper ball out of the open end.

> **Tip:** The arm throw is a first design. Test with a paper ball, then adjust the horn angle.

9 Weather-station box (MC-1, MC-2)



YOU NEED

- MC-1, MC-2
- Arduino UNO R3
- 16×2 LCD, LEDs, buzzers, DHT11
- 4 × M3×6, 4 × M3×10, 4 × M2.5×8 screws

CHECK BEFORE MOVING ON

- USB cable fits out of the side slot.
- No wire is squashed under the lid.

DO THIS

1. Screw the **UNO R3** onto the four round standoffs in **MC-1** with 4 × M3×6 screws. The USB socket lines up with the slot in the left wall.
2. Stick a small breadboard at the back of the box floor with double-sided tape.
3. Screw the LCD under the lid with 4 × M2.5×8, display looking up through the long window.
4. Push the green, blue and red LEDs into the three 5 mm holes and the two buzzers into the two big round holes. A drop of hot glue holds them. Put the yellow LED in the 12 mm hole.
5. Put the DHT11 behind its square window. Connect everything with jumper wires, exactly as in the 3D Wiring Lab (D2 DHT11, D3 green, D4 blue, D5 red, D6 active buzzer, D7 passive buzzer, D8 to D13 LCD, A0 yellow).
6. Close the lid and screw 4 × M3×10 into the four corner posts.

Tip: The 24 mm joystick hole and the button hole are left over from an older design. Cover the big one with tape or a sticker, or leave it open.

10 Earth stand and debris posts (ES-1, ES-2)



YOU NEED

- ES-1, 2 × ES-2
- 70 mm foam ball
- Glue or double-sided tape

CHECK BEFORE MOVING ON

- Ball is held firmly in the ring.

DO THIS

1. Push the foam ball into the ring of **ES-1** (the seat is 70.4 mm wide). A little glue keeps it from rolling off.
2. Put ES-1 in the middle of the board (this is the "Earth" the robot circles).
3. Put the two **ES-2** posts on the board as obstacles, open end down.

Final check and common problems

Problem	Fix
A hole is too tight for a screw or pin	Open it a little with a drill bit by hand, or print again with “horizontal expansion” +0.1 mm in the slicer.
Plate flexes in the middle	Check all 8 strap screws, 8 cradle ear screws and 4 flange screws are in and flush.
Robot leans or one wheel is in the air	Check that all four saddles sit at the same height against the plate and the motors are fully inside.
Robot drifts to one side	Check the left and right motor pairs are wired the same and the wheels are the same size.
Bumper sticks	Clean the pin holes in the guide blocks. Slide the pins in and out a few times.
Hub slips on the stepper shaft	Add a drop of CA glue on the flats of the shaft.

FINAL CHECKLIST

- ☐ Plate is one stiff piece, all screws flush
- ☐ 4 wheels turn freely and touch the table
- ☐ Turret turns, sensor faces forward
- ☐ Camera sees ahead, bumper springs back
- ☐ Cables cannot touch the wheels
- ☐ Tower arm swings and pushes a paper ball
- ☐ Weather box closes with the USB cable outside

Positions of bought parts in the pictures (servo, stepper, modules) are approximate stand-ins. Use the written steps for fitting and test each part before the final screws.